CROSS-MODAL SUBSPACE LEARNING FOR SKETCH-BASED IMAGE RETRIEVAL: A COMPARATIVE STUDY

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Abstract: Sketch-based image retrieval (SBIR) has become a prominent research topic in recent years due to the proliferation of touch screens. The problem is however very challenging for that photos and sketches are inherently modeled in different modalities. Photos are accurate (colored and textured) depictions of the real-world, whereas sketches are highly abstract (black and white) renderings often drawn from human memory. This naturally motivates us to study the effectiveness of various cross-modal retrieval methods in SBIR. However, to the best of our knowledge, all established cross-modal algorithms are designed to traverse the more conventional cross-modal gap of image and text, making their general applicableness to SBIR unclear. In this paper, we design a series of experiments to clearly illustrate circumstances under which cross-modal methods can be best utilized to solve the SBIR problem. More specifically, we choose six state-of-the-art cross-modal subspace learning approaches that were shown to work well on image-text and conduct extensive experiments on a recently released SBIR dataset. Finally, we present detailed comparative analysis of the experimental results and offer insights to benefit future research.

Keywords: SBIR; Cross-modal; Subspace learning; Comparison

1 Introduction

Sketch-based image retrieval (SBIR) has been extensively researched in recent years. Many related datasets [1], [2] and annotations [3] have been recently released. The relationship between sketch and photo is inherently a cross-modal one: sketches are abstract renderings of the visual world, yet photos are exact depictions. This largely motivates us to apply cross-modal retrieval methods to solve the SBIR problem. However, previous cross-modal works mainly focused on image-text datasets (e.g., Pascal VOC dataset [4], Wiki image-text dataset [5]), which makes their general applicableness to SBIR unclear.

The goal of cross-modal techniques is to capture the relevance on an abstract level [4], [5] and shorten the semantic gap between different data sources. Establishing correspondences among data that carry the same semantic definitions yet depicted in different domains (e.g., text/image, and sketch/photo) is commonly refereed as the cross-modal matching problem. As a result, the cross-modal "relevance" is often a highly abstract relationship that can only be learned by constructing a third and shared feature space. This is different to conventional feature presentation learning for single-modality matching where the goal is to find a common subspace for matching. Generally speaking, cross-modal retrieval involves two main tasks: measurement of relevance and selection of coupled features [5]. The challenge of cross-modal matching is therefore to find a semantic feature space that can withstand modal variation at a abstract level. Most cross-modal methods can be classified into three main categories: probabilistic models [6], [7], metric learning approaches [8], [9], and subspace learning methods [10], [11]. Probabilistic approaches aim to model the joint distribution of multi-modal data in order to learn their correlation [6]. Metric learning methods set out to compute appropriate distance metrics between different modalities [8]. Subspace learning approaches map multi-modal data into a common subspace to conduct matching [5]. Among these categories of cross-modal techniques, subspace learning methods have gained state-of-the-art results in recent years [4], [12], [13], [5].

Similar as traditional image-text traversal, it is commonly recognized that once a common subspace is learned between sketch and photo, cross-modal retrieval can be effectively conducted. It is shown that facial photo and sketch matching can be effectively modeled as multi-modal problem via Canonical Correlation Analysis (CCA) [3]. However, no existing work had (i) clearly defined how best photo and sketch matching can be casted in a cross-modal setting, and (ii) spelt out unique characteristics of bridging sketch-photo modalities when compared with the conventional text-image gap.

In this paper, we focus on analyzing the relationship between photo and sketch in the cross-modal setting. In

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978-1-5090-1246-6/16/$31.00 ©2016 IEEE

particular, we benchmark a series of cross-modal subspace learning methods on a recently released SBIR dataset [1], [2]. Through detailed comparative analysis, we offer insights towards the general applicability of cross-modal techniques on matching sketches and photos.

2 Cross photo and sketch modalities

An unique characteristic in the cross-modal problem of sketch-photo, which is fundamentally different to the well-studied problem of text-image, is that both of them are in the visual domain. Despite in the same domain, the semantic gap between sketch and photo is still non-trivial: (i) Photo is an exact digital snapshot of the physical world, maintaining almost all of the original information. Sketches on the other hand are abstract line drawings which often omit many attributes, such as color, brightness, material, texture, etc. Sketch can only capture certain information of shape and contour or surface lines and describe the key features of object, at an abstract and semantic level. (ii) Different people have different observation abilities, drawing styles, and drawing abilities. Sketches drawn by different persons for the same object may capture different information and features due to the gap of drawing styles. As a consequence of the difference in drawing ability, the sketch drawn by a child must be more abstract than the one drawn by an adult. Furthermore, the sketch drawn by a finger may be more abstract than the one drawn by a pen. (iii) Taking text into consideration, photo is perhaps the most concrete in terms of descriptive power where text is the most abstract and subjective. Sketch conveniently sits in the middle being (i) more descriptive than text, yet (ii) more abstract than photo.

3 Cross-modal subspace learning

In this section, all the cross-modal subspace learning methods share a set of notations. Modality a and modality b are p and q dimensional spaces, respectively. Given sample matrices $X_a \in \Re^{p\times n}$ and $X_b \in \Re^{q\times n}$, each pair $\left\{x_i^a, x_i^b\right\}$ represents the same object. We apply Canonical Correlation Analysis (CCA) [4], Bilinear Model (BLM) [14], Partial Least Squares (PLS) [13], Generalized Multi-view Linear Discriminant Analysis [15], Generalized Multi-view Margin Fisher Analysis (GMMFA) [15], and Learning Coupled Feature Space for Cross-modal Matching [5] in SBIR. LCFS, GMLDA, and GMMFA are supervised methods while CCA, PLS, and BLM are unsupervised.

3.1 Canonical Correlation Analysis (CCA)

CCA [4] is a multivariate statistical analysis method similar to principal component analysis (PCA) [16]. It was initially designed for data analysis and dimensionality reduction. Recently, CCA is widely applied in data fusion [4], [17] and has become the most popular unsupervised cross-modal subspace learning method. CCA learns a set of canonical component pairs, i.e., directions $w_a \in \Re^a$ and $w_b \in \Re^b$ along which the data is maximally correlated [4], i.e.,

$$\max_{w_a \neq 0, w_b \neq 0} \frac{w_a^T \sum_{ab} w_b}{\sqrt{w_a^T \sum_{aa} w_a}\sqrt{w_b^T \sum_{bb} w_b}}, \quad (1)$$

where $\sum_{aa}$ and $\sum_{bb}$ represent the empirical covariance matrices for modality a and modality b, respectively, while $\sum_{ab} = \sum_{ba}^T$ represents the cross-covariance matrix between them. Previous work has demonstrated the optimization of (1) can be solved as a generalized eigenvalue problem (GEV) [18].

3.2 Partial Least Squares (PLS)

PLS [19] is usually adopted to perform recognition for feature selection in a wide range of multi-modal scenarios. PLS has been effectively applied for feature selection in face recognition with different motivations [13], [20], [21], [22]. PLS models X_a and X_b such that [13]:

$$X_a^T = TP^T + E, X_b^T = UQ^T + F, U = TD + H. \quad (2)$$

$T \in \Re^{n\times d}$ and $U \in \Re^{n\times d}$ contain the d extracted PLS scores or latent projections. $P \in \Re^{p\times d}$ and $Q \in \Re^{q\times d}$ are the matrices of loadings and $E \in \Re^{n\times p}$, $F \in \Re^{n\times q}$ and $H \in \Re^{n\times d}$ are the residual matrices. $D \in \Re^{d\times d}$ is a diagonal matrix describing the latent scores of X_a and X_b. PLS finds the basis vectors w and c such that the covariance between the score vectors t and u is maximized as

$$\max\left(\left[\operatorname{cov}(t,u)\right]^2\right) = \max\left(\left[\operatorname{cov}(X_a w, X_b c)\right]^2\right) \quad (3)$$
$$s.t. \|w\| = \|c\| = 1.$$

3.3 Generalized multi-view analysis (GMA)

GMA [15] showed us that most popular supervised and unsupervised feature extraction techniques can be derived based on a special framework of a quadratic constrained quadratic program (QCQP), which can be solved efficiently as a generalized eigenvalue problem.

The constrained objective is

$$\left[\hat{v}_1, \hat{v}_2\right] = \arg\max\ v_1^T A_1 v_1 + \mu v_2^T A_2 v_2 + 2\alpha v_1^T Z_1 Z_2^T v_2 \quad (4)$$
$$s.t.\ v_1^T B_1 v_1 + \gamma v_2^T B_2 v_2 = 1,$$

where v_1 and v_2 are the projection directions. The positive terms μ, α, and γ are hyper-parameters for the balances among the objectives. Sharma et al. [15] have illustrated that if we substitute A_i, B_i, and Z_i with

some certain terms in (4), we can obtain the corresponding objective functions of different methods.

1) Bilinear Model (BLM): PCA in the a modality can be formulated as the following eigenvalue problem:

$$X_a W_a X_a^T v_a = \lambda v_a, \quad \textbf{(5)}$$

where $W_a = I_a / n$ and I_a is the identity matrix in the a modality. In (4), setting $A_a = X_a W_a X_a^T$, $B_a = I$, and $Z_a = X_a$, we can get BLM under the GMA framework.

2) Generalized Multi-view Linear Discriminant Analysis (GMLDA): Linear Discriminant Analysis (LDA) in the a modality can be formulated as the eigenvalue problem [15]

$$X_a W_a X_a^T v_a = \lambda X_a D_a X_a^T v_a, \quad \textbf{(6)}$$

where we have $D_a = I - W_a$ [23], [24] ($W_a \in \Re^{n\times n}$, $D_a \in \Re^{n\times n}$). To get GMLDA under the GMA framework, in (4), we can set $A_a = X_a W_a X_a^T$, $B_a = X_a D_a X_a^T$, and $Z_a = X_a$ or $Z_a = M_a$ with M_a defined as the matrix with columns that are class mean values.

3) Generalized Multi-view Margin Fisher Analysis (GMM-FA): Marginal Fisher Analysis (MFA) in the a modality has the following eigenvalue problem:

$$X_a \left(S_{ba} - W_{ba}\right) X_a^T v = \lambda X_a \left(S_{\omega a} - W_{\omega a}\right) X_a^T v, \quad (7)$$

where $S_{(b/w)a}^{kk} = \sum_{kl,k\neq l} W_{(b/w)a}^{kl}$. The within class compression strategy or intrinsic graph in a modality is defined as $W_{\omega a}^{kl}$ [15]. To get GMMFA under the GMA, in (4), we can respectively set A_a, B_a, and Z_a as $X_a \left(S_{ba} - W_{ba}\right) X_a^T$, $X_a \left(S_{\omega a} - W_{\omega a}\right) X_a^T$, and X_a.

3.4 Learning coupled feature spaces for cross-modal matching

Wang et al. [5] has reasonably used that l_{21}-norm has the very powerful capability of feature selection [25], [26], [27] and trace norm [28], [29], [30], [31] can encode the correlation of the design matrix or prior knowledge by performing the corresponding low-rank solution. Hence, they proposed a model in the following form as

$$\begin{aligned} &\min_{U_a,U_b} \frac{1}{2}\left(\left\|X_a^T U_a - Y\right\|_F^2 + \left\|X_b^T U_b - Y\right\|_F^2\right) \\ &+\lambda_1\left(\left\|U_a\right\|_{21} + \left\|U_b\right\|_{21}\right) + \lambda_2 \left\|X_a^T U_a X_b^T U_b\right\|_*, \end{aligned} \quad (8)$$

where U_a and U_b are the projection matrices for the coupled modalities, respectively. The first term is coupled linear regression based on label information, which is used to learn two projection matrices for mapping multi-modal data into a common subspace. The second term containing two l_{21}-norms simultaneously conducts feature selection on two feature spaces. And the trace norm can enforce the relevance of projected data with connections in the subspace.

4 Experimental results and discussions

4.1 Datasets

We perform the aforementioned cross-modal subspace learning methods on a recently released fine-grained sketch- based image retrieval dataset [1], [2]. This shoe dataset has 419 photo-sketch pairs, which can be categorized into three different classes. All the sample pairs are single-labeled. Each shoe photo has a corresponding freehand sketch within various drawing styles. The sketches are drawn by nonexperts using their fingers on a touch screen, therefore these sketches are abstract enough to escape the photo modal.

4.2 Experimental settings

We perform and compare these cross-modal subspace learning methods (including CCA, BLM, PLS, GMLDA, GMM-FA, LCFS) on shoe dataset, for two SBIR retrieval tasks ((1) photos query sketches and (2) sketches query photos). Shoe dataset is so small that the experimental results may contain randomness. Hence, we repeat each experiment on each setting for 50 times and each time we randomly choose 285 sample pairs as training set, and treat the remaining 134 sample pairs as testing set.

In the training phase, we input photo training feature matrix and sketch training feature matrix into these cross-modal retrieval methods to learn a projection matrix for each modality. After model training, we use the projection matrices to map the photo testing samples or sketch testing samples into a common subspace and the cosine distance is adopt to measure the similarity between photos and sketches. Given a photo (or sketch) query, the goal of each SBIR task is to find the nearest neighbors (NN) from the sketch (or photo) database in the aforementioned common subspace. In all the following experiments, we use HOG features extracted from photo modality and sketch modality, respectively. The mean average precision (MAP) [4] is adopted to evaluate the performances of these methods.

4.3 Results on shoe dataset

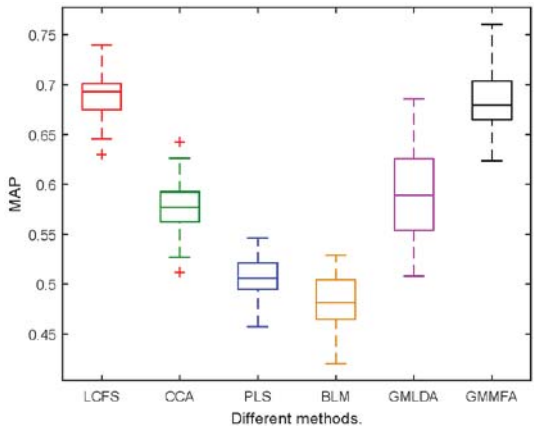
The MAP scores of different cross-modal subspace learning methods are reported in Table I. We repeated each experiment on each setting for 50 times and calculated the minimum (min), maximum (max), mean value (mean), variance (var), and standard deviation (std) for each method, respectively. Experimental results of [5] demonstrated that CCA, PLS, BLM, GMLDA, and

Table I Map scores of different cross-modal subspace learning methods on shoe dataset

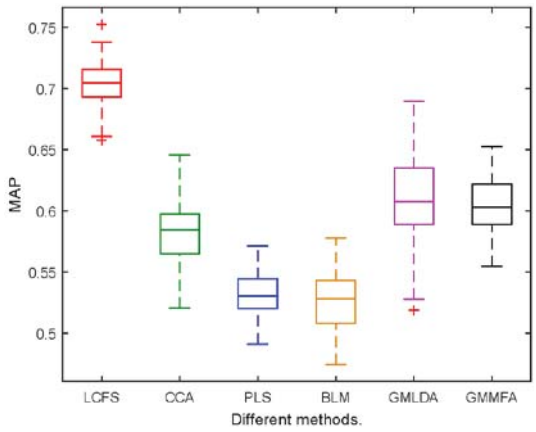
Method	Photo queries sketch					Sketch queries photo				
	min	max	mean	var	std	min	max	mean	var	std
LCFS	0.6142	0.7589	**0.6934**	0.0005	0.0230	0.6327	0.7646	**0.7098**	0.0003	0.0199
PCA+LCFS	0.6086	0.7412	0.6666	0.0006	0.0243	0.6334	0.7601	0.6924	0.0005	0.0230
PCA+CCA	0.5118	0.6425	0.5765	0.0006	0.0257	0.5207	0.6459	0.5820	0.0007	0.0263
PCA+PLS	0.4574	0.5461	0.5076	0.0003	0.0189	0.4914	0.5714	0.5320	0.0004	0.0192
PCA+BLM	0.4188	0.5404	0.4824	0.0006	0.0255	0.4762	0.5730	0.5277	0.0007	0.0268
PCA+GMLDA	0.5082	0.6858	0.5921	0.0023	0.0484	0.5188	0.6897	0.6111	0.0016	0.0401
PCA+GMMFA	0.6133	0.7480	0.6862	0.0010	0.0313	0.5296	0.6724	0.6057	0.0006	0.0241

GMMFA are incapable for feature selection. Thus we utilize Principal Component Analysis (PCA) to refine the original features for these five kinds of methods. In order to validate the coupled feature selection ability of LCFS, hence its results without performing PCA on the original features are also reported, as shown in Table I.

We can observe that: first, LCFS achieves the best performance than other methods for photo querying sketch and sketch querying photo on shoe dataset; second, GMLDA and GMMFA are better than CCA, PLS, and BLM; third, LCFS still has the coupled feature selection ability on sketch while other approaches just focus on the common subspace learning without feature selection. These phenomena are consistent with the results reported in [5].



(a) Photo queries sketch



(b) Sketch queries photo

Figure 1 Box plots of MAP scores for different cross-modal subspace learning methods

All the experimental results are visualized as box plots in Figure 1. The range of certain method shows the performance stability of corresponding method for the SBIR tasks. We can conclude that the performance of GMLDA is most unstable and the remaining methods have similar stabilities. All samples of a dataset are subject to the same underlying data distribution. Theoretically, each method has its own unique principle and its experimental results will be subject to a latent data distribution when it is repeated on the same data set. If we want to know whether the performances of these methods for the SBIR tasks on shoe dataset are fundamentally different, we need to judge whether their experimental result distributions belong to the same distribution or not. In the above discussion, we get a preliminary conclusion that LCFS is the best of these methods for the SBIR tasks on shoe dataset. Moreover, we conducted the student's t-test for the repeated 50 times experimental results between LCFS and other methods, as shown in Table II. The null hypothesis is that the two results have similar means with unknown variance. We can observe that LCFS has the same distribution with GMMFA only for photo querying sketch task. However, their distributions for sketch querying photo are different. In the rest cases, LCFS is significantly different from the others. We can draw a conclusion that the performance of LCFS for the SBIR tasks on shoe dataset is essentially different with the performances of CCA, PLS, BLM, GMLDA, and GMMFA.

Table II P -value comparisons between LCFS and other cross-modal subspace learning methods. The significant level is 0.05

	CCA	PLS	BLM	GMLDA	GMMFA
Photo-Query	6.4e-132	1.4e-65	5.3e-65	1.1e-22	0.2173
Sketch-Query	6.2e-139	2.4e-66	4.2e-60	1.5e-26	1.2e-40
Average-Value	2.6e-137	1.4e-67	4.4e-65	3.3e-25	2.6e-20

LCFS has the best performance in above experiments mainly benefits from two reasons: firstly, it selects the relevant and discriminative features from photo and sketch, and the learned common space is more compact and effective [5]; secondly, it measures correlation and performs feature selection; thirdly, it takes advantage of supervised information by class label. All the terms of its objective are coupled pairs for two modalities, which considered different modalities as an entity. However, all the other methods do not measure correlation and perform feature selection at the same

time. This leads to information losing. GMLDA and GMMFA are better than CCA, PLS and BLM, because the usage of quadratic constrained quadratic optimization are more reasonable by considering photo modality and sketch modality as an entity. Furthermore, GMLDA and GMMFA are supervised methods. They use class information to learn an accurate subspace. CCA and PLS have similar performances because they have similar principles and objective functions. PLS is an improved version of CCA. CCA, PLS and BLM are unsupervised methods. They cannot reach such high accuracy of the supervised methods. These phenomena demonstrate the importance of the supervised information in SBIR.

5 Conclusion

In this paper, we explained the relationship between photo and sketch in different situations to illustrate under what circumstances we can utilize cross-modal methods to solve sketch-based image retrieval (SBIR) problems. We made a brief introduction for a series of cross-modal subspace learning methods and implemented them on a recently released SBIR dataset. We presented detailed comparisons and analysis of experimental results. The cross-modal subspace learning methods can be applied in SBIR tasks. Their performances are almost consistent with those in cross-modal retrieval for image and text. Their ability of feature selection and subspace learning is hardly changed when they were applied in SBIR tasks. LCFS is the best one among the methods compared for both forms of SBIR.

Acknowledgements

This work was partly supported by National Natural Science Foundation of China (NSFC) grant No. 61402047 and 61273217, NSFC-RS joint funding grant Nos. 61511130081 and IE141387, Beijing Natural Science Foundation (BNSF) grant No. 4162044, the Open Projects Program of National Laboratory of Pattern Recognition grant No. 201600018, Scientific Research Foundation for Returned Scholars, Ministry of Education of China, and Chinese 111 program of Advanced Intelligence and Network Service under grant No. B08004.

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